

SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Technical Presentation

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 10%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 50

Code of Conduct

I G. Jaswanth - 192472235 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G. Jaswanth

Assessment Tasks

Technical Presentation Topics

Prepare a 6-8 minute presentation on any one of the following topics. Your slides must include definitions, architecture, industry relevance, and one practical example.

Here are 10 additional technical presentation topics in a similar style, suitable for a 6–8 minute presentation with definitions, architecture, industry relevance, and a practical example:

1. **Containerization and Cloud-Native Applications: Transforming Modern Software Deployment**
 - Explore containers, container architecture, benefits, and a real-world deployment example.
2. **Serverless Computing: Reducing Infrastructure Management in the Cloud**
 - Discuss Function-as-a-Service (FaaS), architecture, business advantages, and practical use cases.
3. **Edge Computing vs Cloud Computing: Choosing the Right Computing Paradigm**
 - Compare architectures, applications, and industry adoption scenarios.
4. **Microservices Architecture in Cloud Environments**
 - Explain microservices design, benefits over monolithic systems, and industry implementations.
5. **Cloud Security Fundamentals: Protecting Data and Applications**
 - Cover security models, shared responsibility architecture, and real-world security practices.
6. **Multi-Cloud and Hybrid Cloud Strategies for Enterprise IT**
 - Analyze deployment models, business benefits, challenges, and practical examples.

7. **The Role of Artificial Intelligence and Machine Learning in Cloud Platforms**
 - Discuss AI/ML cloud services, architecture, and industry applications.
8. **DevOps and Continuous Integration/Continuous Deployment (CI/CD) in Cloud Computing**
 - Explain DevOps principles, CI/CD architecture, and practical implementation examples.
9. **Cloud Storage Technologies: Object, Block, and File Storage Compared**
 - Compare storage architectures, use cases, and industry adoption patterns.
10. **Internet of Things (IoT) and Cloud Computing: Building Connected Ecosystems**
 - Examine IoT-cloud architecture, benefits, challenges, and real-world applications.
11. **Cloud Migration Strategies: Moving Enterprise Applications to the Cloud**
 - Examine migration approaches, cloud architecture considerations, benefits, challenges, and a successful migration case study.
12. **Big Data Analytics in Cloud Computing**
 - Explore cloud-based data processing architectures, analytics platforms, business value, and real-world applications.
13. **Identity and Access Management (IAM) in Cloud Environments**
 - Discuss authentication, authorization architectures, security benefits, and practical enterprise implementations.

Minimum slide flow:

- Title and objective
- Core technical explanation
- Architecture or process diagram
- Real-world use case
- Conclusion and key takeaway
- Limitations and Future Scope

Technical Presentation Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Content	30%	Content is highly accurate, relevant, and insightful	Content is correct with minor gaps	Basic content with limited depth	Content is inaccurate or superficial
Structure and Flow	20%	Excellent logical flow and slide organization	Good structure with minor issues	Some organization but weak flow	Disorganized presentation
Use of Visuals	15%	Visuals strongly support technical communication	Relevant visuals used well	Limited or unclear visuals	No meaningful visuals
Communication Skills	20%	Confident, clear, and engaging delivery	Clear delivery with minor hesitation	Moderate clarity but uneven delivery	Weak delivery and low clarity
Professionalism	15%	Well-prepared and audience-focused presentation	Mostly professional	Basic preparation visible	Poor preparation and professionalism



SIMATS ENGINEERING

**SAVEETHA INSTITUTE OF MEDICAL AND
TECHNICAL SCIENCES**



Identity and Access Management (IAM) in Cloud Environments

Authentication · Authorization · Security Benefits · Enterprise Implementations

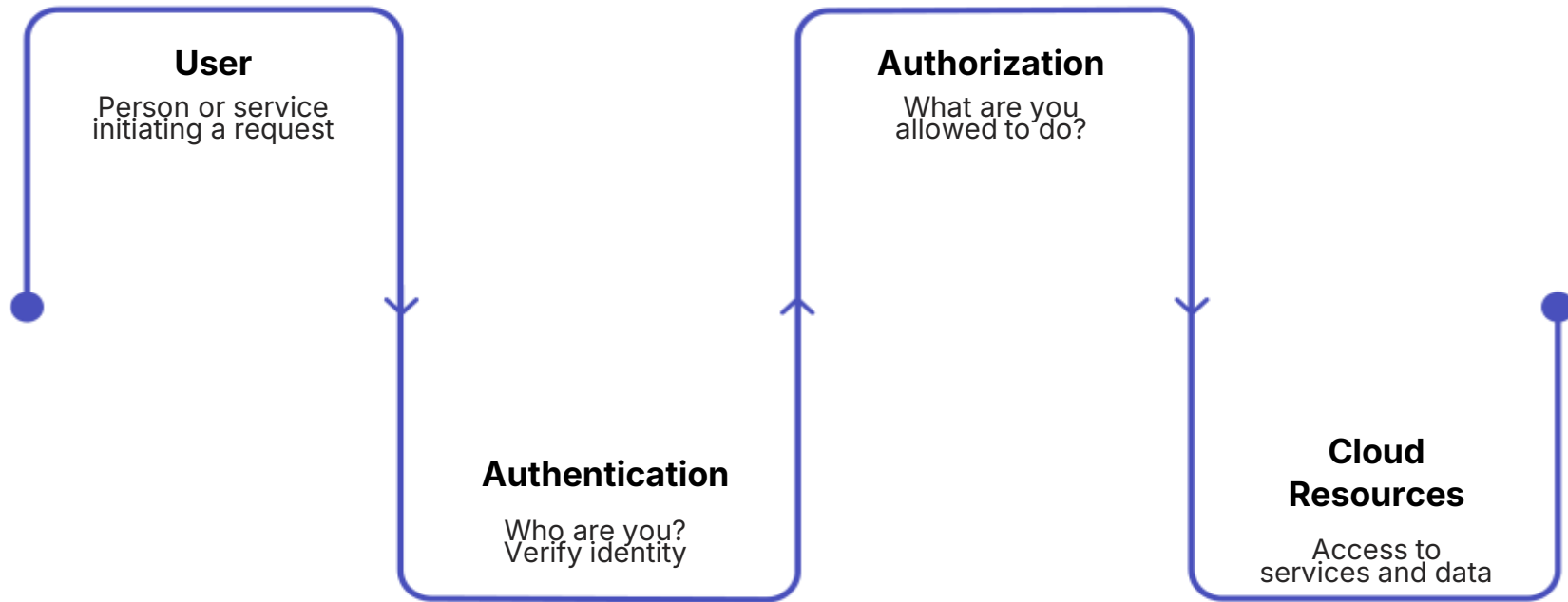
CSA1518 Cloud Computing and Big Data Analytics

Presented by:

G. Jaswanth 192472235

What is IAM?

IAM is a **framework of policies, technologies, and processes** that manages digital identities and controls access to cloud resources. It operates through two core components working in sequence.



Authentication — "Who are you?"

Verifies the identity of a user or system before granting any access.

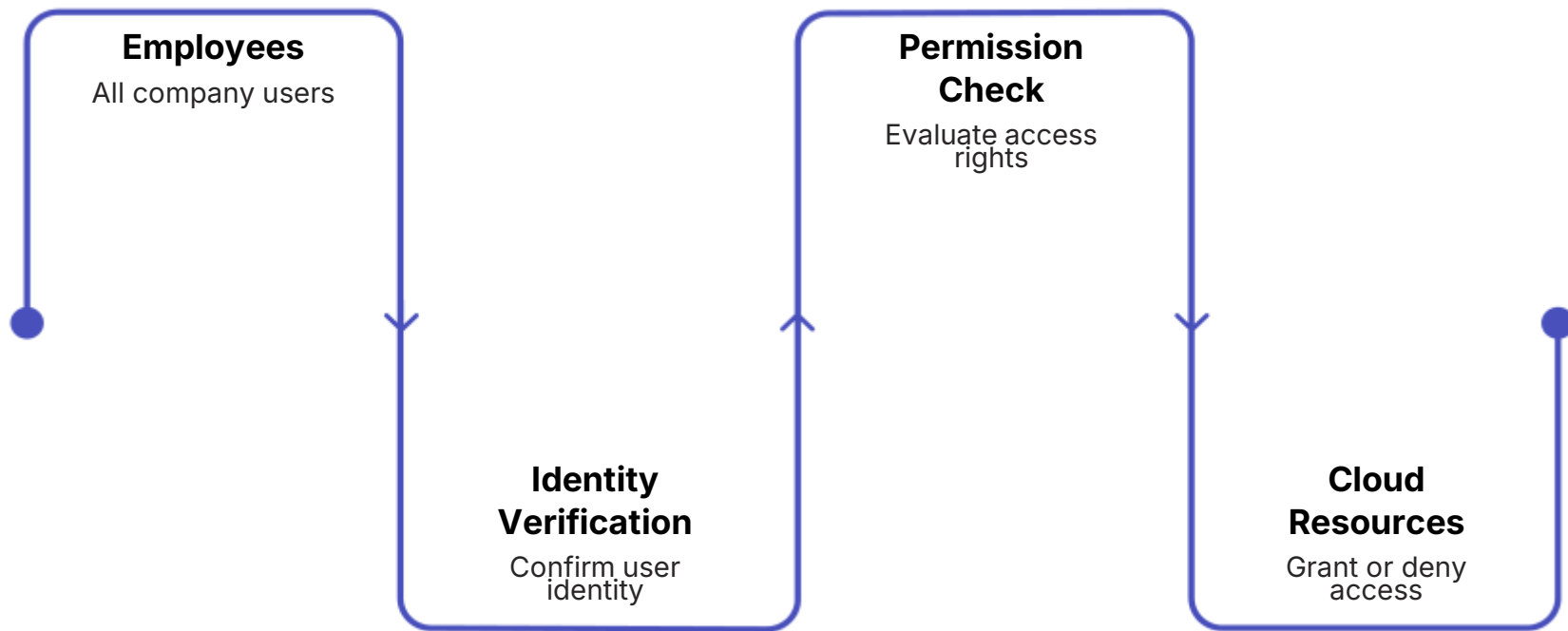
Authorization — "What may you do?"

Determines which resources and actions the verified identity is permitted to use.

Authentication always happens first — the system must know who you are before it can decide what you're allowed to do. These two components are distinct but deeply interconnected in every IAM implementation.

Why Do We Need IAM?

Consider a company with **5,000 employees** across multiple systems — HR Portal, Finance Portal, Cloud Servers, Storage, and Internal Apps. Without IAM, any employee could access any resource



IAM answers: **Who are you?** and **What may you access?** — preventing unauthorized access across every system.

Without IAM, there's no systematic way to control who accesses what. A finance employee could access HR records, a developer could modify production servers, and a terminated employee could retain access indefinitely. IAM solves this by enforcing identity verification and permission checks at every access point.

Authentication

Authentication **verifies identity** before granting access. Passwords alone are no longer sufficient — modern systems use layered verification.



Username & Password

Basic credential-based entry point



OTP

Time-sensitive one-time passcodes



Security Keys

Physical hardware tokens (e.g., YubiKey)



MFA

Multiple verification factors required



Biometrics

Fingerprint, face, or iris recognition



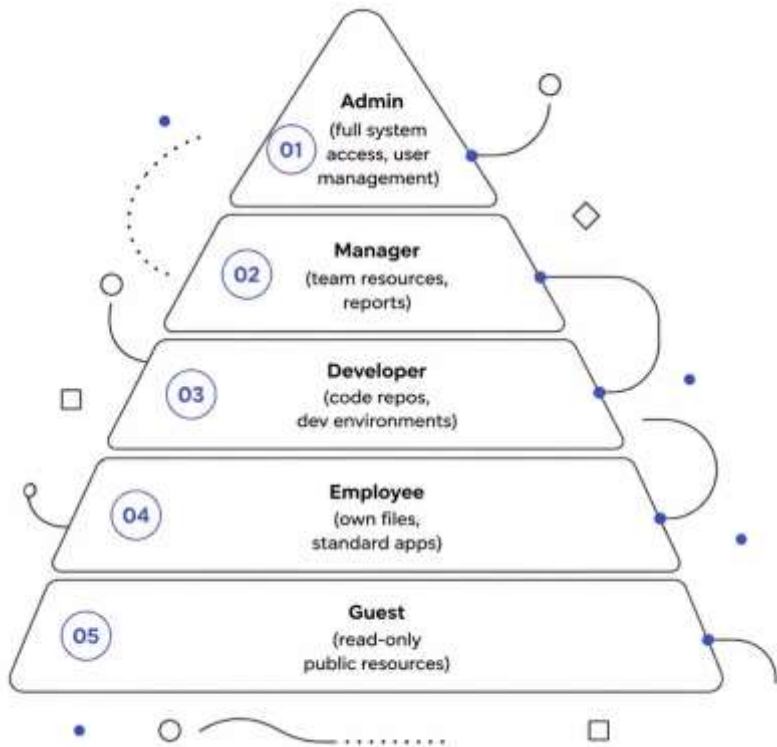
Passwordless

Magic links or push notifications

- ❑ Real-world example: An ATM requires your card (something you have), your PIN (something you know), and sometimes an OTP (something you receive). This is MFA in action. Passwords alone are vulnerable to phishing, brute-force attacks, and credential stuffing — MFA dramatically reduces these risks.

Authorization & Role-Based Access Control

Once authenticated, **authorization** determines what a user may access. **RBAC** assigns permissions based on organizational roles.



Access Decision Workflow

01

Authentication Success

02

Role Evaluation

03

Permission Check

04

Allow or Deny Access

IAM Architecture

A cloud IAM system coordinates multiple components to manage identities and enforce access control at scale.



Component Roles

- **Users** — Employees, systems, or APIs requesting access
- **Authentication Service** — Verifies credentials and MFA factors
- **Identity Provider (IdP)** — Central authority managing digital identities
- **Authorization Engine** — Evaluates policies and roles to grant or deny access
- **Cloud Resources** — Applications, servers, and storage being protected

Security Benefits of IAM



Prevents Unauthorized Access

Only verified identities reach protected resources



Least Privilege Principle

Users receive only the minimum access required



Secure Remote Access

Enforces security policies for distributed workforces



Multi-Factor Authentication

Layered verification reduces credential-based attacks



Audit Logs & Compliance

Full access trails support regulatory requirements



Reduced Insider Threats

Role-based limits minimize damage from compromised accounts

- Each benefit addresses a real-world threat: unauthorized access is blocked at the door, MFA stops credential theft, least privilege limits blast radius, audit logs enable forensics, secure remote access protects distributed teams, and RBAC reduces insider threat impact. Together, they form a comprehensive security posture.

Enterprise IAM Implementations

Major cloud providers implement IAM with different tools but share the same core objective: secure, scalable identity management.

Provider	Core Service	Key Feature	Access Control
AWS	AWS IAM	IAM Roles & Users	JSON Policies
Microsoft	Microsoft Entra ID	Single Sign-On (SSO)	Conditional Access
Google Cloud	Google Cloud IAM	Predefined Roles	Fine-Grained Permissions
Okta	Okta Identity Cloud	Identity Federation	Adaptive MFA & Lifecycle



All providers implement IAM differently but with similar objectives. AWS uses JSON policies for fine-grained control. Microsoft Entra ID (formerly Azure AD) excels at SSO and conditional access. Google Cloud offers predefined and custom roles. Okta specializes in identity federation across multiple platforms. Organizations often combine these tools — for example, using Okta as an IdP with AWS IAM for resource access.

Real-World Case Study: ABC Company

5,000 employees using **Microsoft Entra ID** integrated with **AWS IAM** for unified identity management.



SSO Login

MFA Verify

Role Evaluation

Access Granted

Developer Access

- ✓ Git repositories & dev environments
- ✗ No payroll or HR system access

HR Access

- ✓ Employee records & HR portal
- ✗ No production server access

This demonstrates IAM in a real enterprise: SSO simplifies login, MFA secures it, and RBAC enforces least privilege. Developers can't touch payroll data; HR can't access production servers. Microsoft Entra ID handles identity federation while AWS IAM controls resource-level permissions — a common hybrid architecture.

Best Practices & Key Takeaways

IAM Best Practices



Enable MFA everywhere



Apply least privilege



Use RBAC



Review permissions regularly



Monitor audit logs



Rotate credentials



Adopt passwordless auth

Key Takeaways

Authentication

Verifies identity

Authorization

Grants permissions

IAM Security

Protects cloud resources

Enterprise Scale

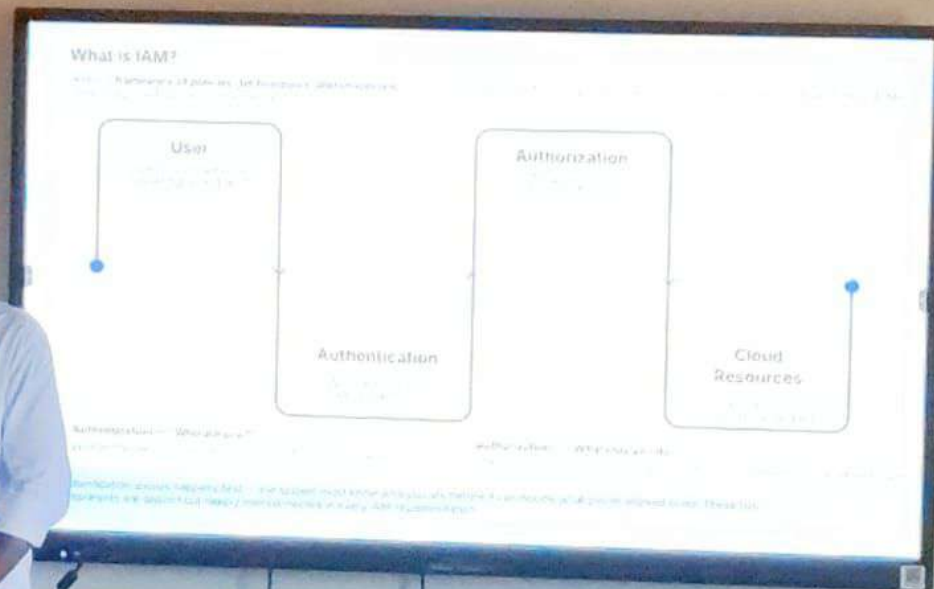
Used by every major cloud provider

Thank You

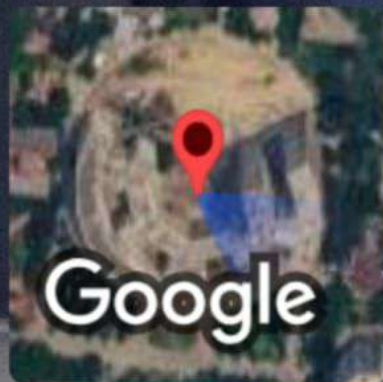
Questions & Discussion



These best practices are industry-standard recommendations from AWS, Microsoft, and Google Cloud security teams. Enable MFA on every account, apply least privilege by default, and review permissions quarterly. Summarize: IAM manages digital identities, authentication verifies who you are, authorization controls what you can do, and together they form the foundation of cloud security.



GPS Map Camera



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